

MAE11 THERMODYNAMICS

Winter 2026

Instructor: Prof. Zahra Sadeghizadeh **Contact Info:** zsadeghizadeh@ucsd.edu

Lectures:

- TR 12:30 PM - 1:50 PM, PCYNH 109
Discussion Session: Friday, 3:00 PM -3:50 PM, PCYNH 106

The discussion sessions will be used for occasional lectures/midterm exam by the professor and as a weekly recitation period where the TAs will work out problems. Attendance is highly recommended for all lectures/recitations. Lectures and recitations are expected to be attended in person.

Recordings will be shared only in the case of documented emergencies and are not available to all students. This option is meant as a short-term accommodation, not a substitute for regular attendance.

Student Hours:

- TR 10:30 AM-11:30 AM
- By appointment. For e-mail to the Instructor or Instructional Assistants, please put “MAE 11” first in the subject box.

Text: Fundamentals of Engineering Thermodynamics M.J. Moran, H.N. Shapiro, D.D. Boettner, M.B. Bailey John Wiley and Sons, Inc, Ninth Edition (eText), 2018.

Course Description: Fundamentals of engineering thermodynamics: energy, work, heat, properties of pure substances, first and second laws for closed systems and control volumes, gas mixtures. Application to engineering systems, power and refrigeration cycles, combustion.

Course Objectives: After taking this class, students will be able to:

- Understand the basic principles of classical thermodynamics.
- Apply principles in analysis of thermodynamic systems.
- Identify engineering applications in energy conversion, refrigeration, and combustion systems, and apply use of thermodynamics in engineering practice, and to promote awareness of impact of engineering solutions
- Develop sound basis for subsequent courses in fluid mechanics, heat transfer, energy technologies and propulsion

Prerequisites: PHYS 2C (Physics—Fluids, Waves, Thermodynamics, and Optics) and CHEM 6A (General Chemistry I) or equivalent are required.

Grading Policy: Homework(12%), class activities, group work, and discussion session activities(8%), Midterm 1 (20%), Midterm 2 (25%), Final (35%).

Important Dates:

Midterm 1	February 5, 2026
Midterm 2	Feb 26, 2026
Final Exam	March 18, 2026
11:30 AM-2:29 PM, MOS 0113	

IA team contact information:

- TA: Mr. Dzafer Camdzic, TA, email: dcamdzic@ucsd.edu, Office hour: TBD
- Reader: Ms. Swetha Chidambara Ganesh, Reader, Email: scg223@ucsd.edu, Office hour: TBD
- Reader: Mr. Thompson Chun Wang Wong, Reader, Reader, Email:tcw003@ucsd.edu, Office hour: TBD
- Reader: Mr. Nicolas J, Liu, Reader, Email: nil015@ucsd.edu, Office hour: TBD

Assessments:

- Your learning in this class will be assessed on an ongoing basis using in class group work activities, midterm exam, and at the end of the quarter with a traditional final exam.
- **Class participation:** is measured by in-class and discussion session group work or individual activities. This cannot be made-up for any reason. Attendance is mandatory to receive credit for class activities, which are an integral part of the learning experience. Class activities will be accessible only during the lecture and for 30 minutes after the session ends. The two lowest scores will be dropped.
- **HWs:** HWs are an integral part of the learning process as they are meant to generate questions and to allow you to acquire a deeper understanding of the important concepts covered in this class. HWs must be your OWN work. Do not copy off OTHER (AI Tools, friends, solution manuals, books etc.) sources. Ditto for exams. Any copying as defined above will be grounds for a F grade. Lowest HW grade will be dropped. HWs will be submitted on Gradescope.
- Questions regarding your grade on the homework assignments or midterms will be accepted only within 48 hours after the grade is released.
- Asking questions during the lectures is encouraged and appreciated.
- Exams are closed-book/notes, and are not cumulative.
- Note 1: There will be no make-up exam for the midterms.
- Note 2: Missed final exam will result in an Incomplete grade and requires doctor/hospital documents for a make-up final during the Final Exams week of the quarter.

Academic Integrity: Students are expected to uphold the highest standards of academic integrity and engineering professionalism. Collaboration with classmates is encouraged for learning purposes, such as discussing concepts or general problem-solving approaches. However, all submitted work must represent each student's own independent effort. Unacceptable collaboration includes copying or sharing solutions, submitting substantially similar work, or dividing homework problems among students. Each student must complete and submit their own homework, quizzes, and exams. The use of AI tools or other online solution resources is not permitted for homework, quizzes, or exams. Using AI to generate or guide solutions is considered an unauthorized resource. Violations of academic integrity will result in academic penalties and may be reported in accordance with the UC San Diego Policy on Academic Integrity: <https://students.ucsd.edu/academics/academic-integrity/>

Accommodations: Students requesting special accommodations for the course and exams must provide a current Authorization for Accommodation (AFA) letter from the Office for Students with Disabilities (OSD). This letter must be emailed to the Instructor by Friday, January 9, 2026.

Week	Topics & Homework
1, Jan 5	Course Information, Introduction Chapter 1. Basic Concepts Chapter 2. Energy, Work, Heat, First Law (closed System)
2, Jan 12	Chapter 2. First Law (closed System) Chapter 3. Properties of pure substances, Ideal Gas, Incompressible Substance HW 1 collected
3, Jan 19	Chapter 3. Properties of pure substances, Ideal Gas, Incompressible Substance Chapter 4. First Law (control volume) HW 2 collected review for exam
4, Jan 26	Chapter 4. First Law (control volume) HW 3 collected
5, Feb 2	Chapter 5. Introduction to Second Law, Carnot Cycle, Temperature Scale Midterm 1 on Feb 5 HW 4 collected
6, Feb 9	Chapter 6. Entropy Chapter 6. Second Law (closed system) HW 5 collected
7, Feb 16	Chapter 8. Cycle Analysis - Vapor Power Cycles
8, Feb 23	Chapter 10. Refrigeration Cycles Midterm 2 on Feb 26 HW 6 collected
9, Mar 2	Chapter 9. Gas Power Cycles
10, Mar 9	Chapter 9. Gas Power Cycles Final Exam Review and HW 7 collected

NOTE: The above tentative course content and actual dates may be modified due to the requirements of the class. Also, the indicated dates may be moved backward or forward depending on class progress.